ZS-S2

Neutrino Mass Spectrum & HNL Phenomenology: The "33 GeV Ghost" — Seesaw Parameters, Experimental Bounds, and BBN Safety

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Verification: 25/25 PASS | Zero Free Parameters

§0. Abstract

Type-I seesaw with m_D = m_e × A = 40.93 keV (ZS-M2 v1.0 transduction, zero free parameters) predicts a heavy neutral lepton (HNL) at M_R = 33.50 GeV. The seesaw mixing angle $|\theta|^2 = 1.49 \times 10^{-12}$ is 10^6–10^7 times below all current and projected experimental bounds (LEP, LHC, SHiP, FCC-ee). The HNL decays with lifetime τ = 38 ns ≪ t_BBN, ensuring Big Bang Nucleosynthesis safety. The Z_2 flavor symmetry from ZS-F5 v1.0's $\hat{W}^2=I$ forces $M_2=M_3$ at tree level, motivating μ–τ reflection symmetry ($\theta_{23}=45°$, δ_CP=±π/2) and the quasi-degenerate spectrum required for resonant leptogenesis (see ZS-S5 v1.0). Davidson–Ibarra exclusion confirms hierarchical thermal leptogenesis is impossible at M_R = 33.5 GeV; only resonant/ARS mechanisms (ZS-S5 v1.0) are compatible. All results use zero new free parameters beyond A = 35/437. Verification: 25/25 PASS.

Keywords: seesaw mechanism, heavy neutral lepton, HNL phenomenology, neutrino mass, μ–τ symmetry, BBN safety, mixing angle, Z_2 flavor symmetry

Epistemic Status Legend

Status	Definition
PROVEN	Follows from standard mathematics alone. Machine-verifiable.
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
DERIVED-CONDITIONAL	Derived from Z-Spin axioms, conditional on a stated assumption.
VERIFIED	Numerically confirmed against observational data or independent computation.
TESTABLE	Well-defined prediction awaiting experimental data.
HYPOTHESIS	Physically motivated conjecture. Derivation chain incomplete.
STANDARD	Accepted SM/cosmological input (not Z-Spin specific).
CONSISTENT	Compatible with framework structure but not independently derived.
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.
OPEN	Well-posed problem without current resolution.
RETRACTED	Previously claimed, now withdrawn with documented reason.

§1. Introduction

The neutrino sector remains one of the most active frontiers in particle physics. Neutrino oscillation experiments have established non-zero masses, but the Standard Model provides no mechanism for generating them. The Type-I seesaw mechanism introduces right-handed neutrinos with Majorana masses M_R, generating light neutrino masses $m_\nu = m_D^2/M_R$. In conventional seesaw models, M_R is a free parameter typically placed near the GUT scale ($\sim 10^1$–10^{15} GeV), far beyond experimental reach.

Z-Spin Cosmology eliminates this freedom entirely. The Dirac mass $m_D = m_e \times A$ arises from the Z-seam transduction mechanism (ZS-M2 v1.0), fixing $M_R = m_D^2/m_{atm} = 33.50$ GeV with zero adjustable parameters. This places the heavy neutral lepton (HNL) squarely within the electroweak energy range — kinematically accessible at LEP and LHC, yet rendered invisible by the seesaw mixing suppression $|\theta|^2 = 1.49 \times 10^{-12}$. The result is a "33 GeV Ghost": a particle predicted at a specific mass with specific couplings, yet undetectable by any foreseeable experiment.

Scope Declaration. ZS-S2 is the CANONICAL source for all Z-Spin neutrino sector parameters: seesaw mass hierarchy, mixing angle, HNL lifetime, Z_2 flavor symmetry, and direct experimental search bounds. For the dynamical leptogenesis mechanism see ZS-S5 v1.0. For the QKE kernel-level microphysics closure, see ZS-U7 v1.0 Appendix A.

§2. Locked Inputs

Parameter	Value	Source	Status
A	35/437 = 0.08009	ZS-F2 v1.0	**LOCKED**
m_e	0.511 MeV	PDG 2024	**STANDARD**
m_{atm} ($\sqrt{\Delta m^2_{atm}}$)	0.050 eV	NuFIT 5.2	**STANDARD**
v_{EW}	246.22 GeV	PDG 2024	**STANDARD**
G_F	1.166×10^{-5} GeV^{-2}	PDG 2024	**STANDARD**

Note on m_{atm}: NuFIT 5.2 central value for normal ordering is $\sqrt{\Delta m^2_{31}} = 0.0501$ eV. The canonical chain uses the rounded value 0.050 eV for cross-paper consistency, yielding $M_R = 33.50$ GeV. The difference ($\delta m_{atm}/m_{atm} = 0.2\%$) propagates to $\delta M_R/M_R = 0.4\%$, well within the seesaw approximation uncertainty.
Note on mass ordering (April 2026 addendum): The numerical value $m_{atm} = \sqrt{|\Delta m^2_{31}|} \approx 0.050$ eV is independent of the mass ordering assumption (NO or IO), since $|\Delta m^2_{31}|$ is the same observable in both. ZS-S2 v1.0 inherited the NuFIT 5.2 NO labelling for historical reasons. ZS-Q5 v1.0 §4.1a establishes the contragredient branch selection $\delta_{CP} = -\pi/2 - \arctan(A) = 265.42°$, which matches the NuFIT 6.0 IO best-fit (~270°) at 0.23σ and selects Inverted Ordering as the Z-Spin canonical prediction. The Book §27.4 mirrors this choice. ZS-S2 §3.1 (added in this April 2026 update) makes the NO/IO assignment of N_1, N_2, N_3 to (m_1, m_2, m_3) explicit and shows that the seesaw chain $m_D = m_e \times A \rightarrow M_R = 33.5$ GeV survives unchanged in the IO interpretation. NuFIT 6.0 reference: Esteban et al., JHEP 12 (2024) 216.

§3. Seesaw Mass Hierarchy (Canonical Derivation)

The Type-I seesaw with ZS-M2 v1.0 transduction yields:

$m_D = m_e \times A = 0.511\ MeV \times (35/437) = 40.93\ keV$ (1)

$M_R = m_D^2/m_atm = (40.93\ keV)^2/(0.050\ eV) = 33.50\ GeV$ (2)

$Y_0^2 = 2 \cdot m_atm \cdot M_R/v^2_EW = 5.53 \times 10^{-14}$ (3)

Self-consistency: $m_\nu = m_D^2/M_R = m_atm$ exactly. Zero free parameters.

[STATUS: DERIVED] *All from ZS-F2 v1.0 (A) + ZS-M2 v1.0 (transduction) + standard seesaw. No tuning.*

Physical origin of m_D = m_e × A:

The choice of reference mass m_e (not m_μ, m_τ, or quark masses) follows from the SU(2)_L doublet structure: ν_L and e_L share the same Yukawa root in the SM Lagrangian. The suppression factor A arises from Z-seam mediation, which is forced by ν_R's U(1)_Y = 0 (SM gauge singlet). In Z-Spin language, U(1)_Y is the Z-sector (ZS-S1 v1.0 §8.2); therefore ν_R is Z-sector decoupled, and its Yukawa coupling to the Higgs must be mediated through the Z-seam with transduction amplitude A (TH-1, topological DoS linearity).

[STATUS: DERIVED under TH-1 + SU(2)_L Hypercharge Argument] *Open Problem TH-1-OPN: Prove Z-seam transduction amplitude = A from canonical topological field theory. Kill condition: if amplitude ∝ A^n (n ≠ 1) then m_D = m_e × A^n and M_R shifts accordingly.*

§3.1 Mass Ordering Assignment (April 2026 addendum)

ZS-Q5 v1.0 §4.1a (contragredient branch selection) and the Book §27.4 establish Inverted Ordering (IO) as the Z-Spin canonical mass ordering prediction. ZS-S2 v1.0 was written under the implicit Normal Ordering convention inherited from NuFIT 5.2; this subsection makes the IO assignment of (N_1, N_2, N_3) to the light neutrino mass eigenstates (m_1, m_2, m_3) explicit, and verifies that the canonical seesaw chain m_D = m_e × A → M_R = 33.50 GeV survives unchanged in the IO interpretation.

IO assignment (canonical,): The Z_2-degenerate HNL pair (N_2, N_3) at M_pair = 33.50 GeV generates the two heavy light-neutrino mass eigenstates (m_1, m_2), with $m_1 \approx m_2 \approx \sqrt{|\Delta m^2_{31}|} \approx 0.0492$–$0.0500$ eV (quasi-degenerate at the "atmospheric" scale). The Z_2-singlet HNL N_1 (eigenvector of U = diag(+1,−1,−1) with eigenvalue +1) corresponds to the lightest mass eigenstate $m_3 \approx 0$. The small solar splitting $m_2 - m_1 \approx 7.4 \times 10^{-4}$ eV arises from Z_2-breaking spurion corrections, controlled by A through the same mechanism as the resonance gap reduction (ZS-S5 v1.0 §4 + ZS-S4 v1.0 Texture Zero Lemma).

Seesaw chain invariance: Equations (1)–(3) of §3 use $m_atm = \sqrt{|\Delta m^2_{31}|}$ as input, and $|\Delta m^2_{31}|$ is the SAME observable in both NO and IO. The numerical values m_D = 40.93 keV, M_R = 33.50 GeV, $|\theta|^2 = 1.49 \times 10^{-12}$, τ_N = 38 ns, $Y_0^2 = 5.53 \times 10^{-14}$ are unchanged under the IO reinterpretation. Only the assignment of N_i to m_i is updated.

Cosmological prediction: $\Sigma m_\nu \approx m_1 + m_2 + m_3 \approx 0.0992$ eV (with $m_3 \approx 0$). PASSES Planck18+BAO bound (< 0.12 eV) with 17% margin. PRESSURED by DESI strict full-shape bound (< 0.071 eV) — see new gate F-S2-IO1 in §8.1. The effective Majorana mass for $0\nu\beta\beta$ in IO has a structural floor $m_{\beta\beta} \approx$ 0.015–0.050 eV depending on Majorana phases — see new gate F-S2-IO2.

[STATUS: DERIVED-CONDITIONAL] *Conditional on ZS-Q5 v1.0 §4.1a contragredient branch derivation (NC-Q5.2: rigorous QFT derivation pending). NuFIT 6.0 currently prefers NO at ~2.7σ; IO assignment is in tension at this level but not excluded. Mass ordering will be resolved by JUNO (~2027) and DUNE (~2030) — registered as gate F-Q5.8 (ZS-Q5 v1.0 §9), cross-linked here as F-S2-IO0 in §8.1.*

§4. The "33 GeV Ghost": HNL Visibility

4.1 Seesaw Mixing Angle

$$\theta \approx m_D / M_R = (40.93 \text{ keV}) / (33.50 \text{ GeV}) = 1.22 \times 10^{-6} \quad (4)$$

$$|\theta|^2 = 1.49 \times 10^{-12} \quad (5)$$

This suppresses ALL HNL interactions with SM gauge bosons by $\sim 10^{-12}$. The HNL effectively decouples from the visible sector.

[STATUS: DERIVED] *$\theta = m_D/M_R$ from seesaw with $m_D = m_e \times A$ locked.*

4.2 Direct Search Bounds

Experiment	Channel	M (GeV)	$\lvert\theta\rvert^2$ Limit	Z-Spin Ratio	Reference
DELPHI (LEP)	$Z \to \nu N$	33	$< 10^{-5}$	10^{-7}	[10]
L3 (LEP)	$Z \to \nu N \to \nu q\bar{q}\ell$	33	$< 2\times10^{-5}$	7×10^{-8}	[11]
CMS (13 TeV)	$pp \to W^* \to \ell N$	30	$< 10^{-5}$	10^{-7}	[12]
ATLAS (13 TeV)	$pp \to W^* \to \ell N$	30	$< 5\times10^{-6}$	3×10^{-7}	[13]
FCC-ee (proj.)	$Z \to \nu N$	33	$\sim10^{-8}$	10^{-4}	[14]
SHiP (proj.)	Charm → N	~30	$\sim10^{-9}$	10^{-3}	[15]

All bounds are 10^6–10^7 above the Z-Spin prediction. Even next-generation experiments (FCC-ee, SHiP) remain 10^3–$10^4\times$ above $|\theta|^2 = 1.49 \times 10^{-12}$.

[STATUS: DERIVED] *The Z-Spin HNL at 33 GeV is a true "phantom": predicted but undetectable by any foreseeable direct search.*

4.3 LEP Invisible Width

Since $M_R = 33.5$ GeV $< M_Z/2 = 45.6$ GeV, the decay $Z \to \nu N$ is kinematically allowed:

$\Gamma(Z\to\nu N) = \Gamma(Z\to\nu\bar{\nu})_SM \times |\theta|^2 \times (1-x)^2(1+x/2)$ (6)

where x = $(M_R/M_Z)^2 = 0.135$. Phase space factor = 0.799. Signal (3 gen): 6.0×10^{-4} eV. LEP precision: ±1.5 MeV. Ratio = 4×10^{-10}. Invisible at LEP.

[STATUS: DERIVED] *LEP Z-width constraint satisfied by 10 orders of magnitude.*

§5. BBN Safety

5.1 Decay Rate Calculation

For M_R < M_W, the HNL decays via off-shell W and Z bosons through three-body channels:

$\Gamma_N = (G_F^2 M_R^5 / 192\pi^3) \times |\theta|^2 \times N_eff$ (7)

where N_eff ≈ 11.9 accounts for all kinematically accessible CC and NC channels (Atre et al. 2009 [9]):

$G_F^2 M_R^5 / 192\pi^3 = 964$ eV, $|\theta|^2 = 1.49 \times 10^{-12}$, N_eff = 11.9

$\Gamma_N = 1.71 \times 10^{-8}$ eV ⇒ $\tau_N = \hbar/\Gamma_N = 3.84 \times 10^{-8}$ s ≈ 38 ns (8)

[STATUS: DERIVED] *τ_N = 38 ns from locked parameters. No free parameters.*

5.2 Comparison with BBN Timescales

$\tau_N / t_BBN = 3.84 \times 10^{-7}$ (2.6×10^6 times shorter than BBN onset).

5.3 Thermal History

At T ~ M_R ~ 33 GeV, the HNL production rate via all scattering channels (CC + NC + inverse decay) exceeds the Hubble rate: Γ_prod/H ≫ 1 (thermalized). Detailed thermal field theory computations including all 2→2 scattering processes with SM fermions in the bath confirm robust thermalization at this temperature. This is beneficial for leptogenesis (ZS-S5 v1.0 §4): it provides the initial N_2,N_3 population. At T_BBN ~ 1 MeV: exp(−M_R/T_BBN) = exp(−33500) ≈ 0. No HNL survives to BBN.

[STATUS: DERIVED] *Thermalized at T ~ M_R, τ = 38 ns ≪ t_BBN. ZS-U4 v1.0 BBN predictions unaffected.*

§6. Z_2 Flavor Symmetry & μ–τ Reflection

$P_{\mu\tau}\, M_\nu\, P_{\mu\tau} = M_\nu^*$ (9)

Chain: $\hat{W}^2=I$ (ZS-F5 v1.0) → κ=4 witness → P_{μτ} → $M_2=M_3$ at tree level. The Z_2 seam involution $\hat{W}$ acts as an order-2 element on the Q=11 slot register. The κ=4 witness projects this involution onto the

3-generation flavor space, where it acts as the μ–τ permutation matrix P_{μτ}. This is a group-theoretic consequence, not an assumption.

[STATUS: PROVEN] *From ZS-F5 v1.0 seam involution. Exact degeneracy at A=0.*

Explicit involution and N_1 Z_2-singlet identification (April 2026 addendum): The κ=4 witness projects the seam involution Ŵ onto the three HNL species (N_1, N_2, N_3) as the diagonal operator U = diag(+1, −1, −1). This is the precise group-theoretic content of the μ–τ reflection action: N_1 is the +1 eigenvector (Z_2-singlet, dim = 1), and (N_2, N_3) span the −1 eigenspace (Z_2-doublet, dim = 2). The Z_2-invariance condition U M_R U = M_R forces $(M_R)_{22} = (M_R)_{33} \equiv$ M_pair, but does NOT constrain $(M_R)_{11}$ — N_1 and the (N_2, N_3) pair live in different irreducible representations of the Z_2. The canonical seesaw chain (§3) determines M_pair = 33.50 GeV; the Z_2-singlet mass $(M_R)_{11}$ is not pinned by §3 alone. The Book §7.3 records this structural fact.

[STATUS: PROVEN] *U = diag(+1, −1, −1) follows directly from $\hat{W}^2$=I and the κ=4 witness as a representation on the 3-generation HNL space. Implication for §3.1: in the canonical IO assignment, (N_2, N_3) generate the quasi-degenerate (m_1, m_2) at the atmospheric scale, and N_1 corresponds to $m_3 \approx 0$.*

N_1 Yukawa structure (DERIVED-CONDITIONAL, April 2026 update): The IO scenario requires $m_3 \approx 0$, which is realised structurally by m_{D,1} = 0 (N_1 decoupled from the SM lepton doublets at tree level) — the "minimal seesaw" structure of Frampton–Glashow–Yanagida (2002). The derivation chain is now closed by character orthogonality of the icosahedral group I ≅ A_5: (i) ZS-M9 v1.0 §3 Table 2 assigns ν_R to the trivial irreducible representation 1 of I ≅ A_5 (HYPOTHESIS strong, 5 lines of evidence); (ii) under this assignment, the Yukawa coupling Y^ν L̄ H ν_R^{(1)} lives in 3 ⊗ 5 ⊗ 1, where L is in irrep 3 and H is in irrep 5; (iii) ZS-M11 v1.0 §9.5 Theorem 9.5.1 (April 2026 update) proves dim Hom_I(1, 3 ⊗ 5 ⊗ 1) = ⟨χ_3, χ_5⟩ = 0 by character orthogonality (3 ≠ 5 are distinct irreducible representations of I); (iv) therefore Y^ν_{α1} = 0 identically by I-symmetry, hence m_{D,1} = 0 and m_3 = 0 in the I-symmetric limit, regardless of $(M_R)_{11}$. ZS-M11 v1.0 §9.5.2 further shows that among all five irreducible representations of I ≅ A_5, the trivial irrep 1 is uniquely the irrep that forbids the Yukawa coupling 3 ⊗ 5 ⊗ X by character orthogonality — the ZS-M9 Table 2 assignment is therefore precisely the assignment that produces a decoupled (massless tree-level) right-handed neutrino species. Open Problem TH-1-OPN (§3) is hereby narrowed: the "m_{D,1} = 0 mechanism" sub-question is resolved (PROVEN at character level); only the upgrade of the ZS-M9 Table 2 assignment from HYPOTHESIS strong to PROVEN remains open. [STATUS: DERIVED-CONDITIONAL. The character calculation dim Hom_I(1, 3 ⊗ 5 ⊗ 1) = 0 is PROVEN unconditionally (ZS-M11 v1.0 §9.5.1, April 2026 update). The physical consequence m_{D,1} = 0 is conditional only on the ZS-M9 v1.0 §3 Table 2 assignment ν_R ⟷ I-irrep 1, which retains HYPOTHESIS strong status. If that assignment is upgraded to PROVEN by future work, m_{D,1} = 0 is upgraded to PROVEN automatically.]

When A ≠ 0 breaks Z_2 → Z'1: radiative corrections lift degeneracy. Mass splitting hierarchy:

A-controlled ΔM^(A) ~ $O(10^{-15})$ GeV dominates charged-lepton ΔM^(W) ~ $O(10^{-19})$ GeV by ~10^3×.

Implications for the resonance condition (ΔM/Γ) and leptogenesis viability: see ZS-S5 v1.0 §4.

The μ–τ reflection symmetry predicts $\theta_{23} = 45°$ and $\delta_CP = \pm\pi/2$ ($|\sin\delta| = 1$). The derivation chain is: $\hat{W}^2=I$ (ZS-F5 v1.0) → κ=4 → $P_\{\mu\tau\}$ → $\delta_CP = \pm\pi/2$ → $|\sin\delta| = 1$. This upgrades sinφ from ASSUMPTION to DERIVED (ZS-M5 v1.0).

§7. Leptogenesis Compatibility (Scope Note)

7.1 Davidson–Ibarra Exclusion

$$|\varepsilon_1| \leq (3/16\pi) \times (M_1/v^2) \times m_3 = 1.65 \times 10^{-15} \quad (10)$$

Required asymmetry: $\varepsilon_req \sim 1.8 \times 10^{-5}$. The DI bound is $\sim 10^{10}$ too small. Standard thermal leptogenesis is impossible at M_R = 33.5 GeV.

[STATUS: DERIVED] *DI bound excludes hierarchical leptogenesis. Only resonant/ARS compatible.*

7.2 Resonance Gap & Resolution Pathway (→ ZS-S5 v1.0)

Naive radiative splitting: $\Delta M_rad \sim (A^2/16\pi^2) \times M_R \approx 1.36$ MeV. Gap: $\Delta M_rad/\Gamma_N \sim 8 \times 10^{13}$ (OPEN).

ZS-S4 v1.0's Z_2 Texture Zero Lemma [PROVEN] reduces gap to $\Delta M/\Gamma \sim 163$ (PARTIALLY RESOLVED).

Full resonant/ARS leptogenesis framework, Pilaftsis–Underwood mechanism, Casas–Ibarra parametrization, spectator chemistry, and Boltzmann equations: see ZS-S5 v1.0.

QKE kernel-level closure with finite-rate sphalerons and c_sph = 28/79 derivation: see ZS-U7 v1.0 Appendix A.

§8. Falsification Conditions

Gate	Condition	Experiment	Status
FS2-1	$\|\theta\|^2 > 10^{-5}$ at 33 GeV (>5σ)	FCC-ee, SHiP	**TESTABLE**
FS2-2	Y_p or D/H incompatible with M_R=33 GeV HNL	AlterBBN	**TESTABLE**
FS2-3	4th light ν confirmed (>5σ)	Short-baseline	**TESTABLE**
FS2-4	M_R > T_sph = 131.7 GeV (sphalerons inactive)	Lattice EW	**DERIVED**
FS2-5	m_D/m_e deviates from A by >5%	Theoretical	**DERIVED**

§8.1 IO Consistency Gates (April 2026 addendum)

The five gates FS2-1 through FS2-5 (§8) remain active and unchanged. The following gates are added in this April 2026 update to falsify the IO assignment introduced in §3.1 and §6 (April 2026 addenda). Gate

F-S2-IO0 is a cross-link to the existing ZS-Q5 v1.0 gate F-Q5.8; gates F-S2-IO1 and F-S2-IO2 are new external-data gates; F-S2-IO3 is a theoretical (derivation-required) gate. All gates use the standard 3σ falsification threshold.

F-S2-IO0 (cross-link to F-Q5.8): If JUNO confirms Normal Ordering at >5σ, the IO assignment in §3.1 and §6 is falsified. Experiment: JUNO (~2027), DUNE (~2030), Hyper-Kamiokande. Status: TESTABLE. Current state: NuFIT 6.0 prefers NO at ~2.7σ; below the 5σ threshold. Note: this gate falsifies the IO ASSIGNMENT only, not the seesaw chain $m_D = m_e \times A \rightarrow M_R = 33.50$ GeV, which holds in both NO and IO interpretations.

F-S2-IO1 (cosmological neutrino mass): If Σm_ν is established below 0.085 eV at >3σ (excluding the IO floor at ~0.099 eV), the IO assignment is falsified. If Σm_ν is established above 0.105 eV at >3σ, both NO and IO are challenged jointly (a different problem). Experiment: DESI DR3 + Euclid + CMB-S4 + Planck. Status: TESTABLE. Current state: Planck18+BAO < 0.12 eV (PASS); DESI strict full-shape < 0.071 eV (TENSION; depends on systematic checks).

F-S2-IO2 (effective Majorana mass $m_{\beta\beta}$): If 0νββ experiments robustly exclude $m_{\beta\beta} > 0.013$ eV at >3σ across the entire IO Majorana-phase parameter space, the IO assignment with $m_3 \approx 0$ is falsified. Experiment: KamLAND-Zen, LEGEND-1000, nEXO (~2030). Status: TESTABLE. Current state: KamLAND-Zen < 0.036 eV (PASS); LEGEND-1000 projected sensitivity ~0.012–0.020 eV will be decisive.

F-S2-IO3 (theoretical, Yukawa-side spurion derivation): [Revised, April 2026 update.] The original v1.1.0 conjecture — that the solar splitting Δm^2_{21} and the leptogenesis resonance gap (F23-R1) share a common Majorana-side spurion mechanism — is FALSIFIED at the order-of-magnitude level. The required Majorana mass splitting for the IO solar splitting is $\Delta M_{pair} \approx (m_2 - m_1) \times M_{pair}^2 / m_D^2 \approx 0.5$ GeV (about 1.5% of $M_{pair} = 33.5$ GeV), whereas the bounded Majorana spurion of Paper 24 §4 (the leptogenesis resonance gap, $\Delta M / \Gamma_N \sim 163$ with $\Gamma_N \approx 1.73 \times 10^{-14}$ GeV) gives $\Delta M_{resonance} \approx 3 \times 10^{-15}$ GeV. The ratio $\Delta M_{pair} / \Delta M_{resonance} \approx 10^{15}$ cannot be reconciled by any single common spurion: the two splittings are 14 orders of magnitude apart and must originate from independent breaking mechanisms. Corrected picture: the Majorana side preserves exact Z_2-degeneracy ($M_{22} = M_{33} = 33.5$ GeV), satisfying the leptogenesis resonance gate F23-R1 with $\Delta M_R \rightarrow 0$; the Yukawa side carries a small spurion ε in $m_{D,2}$ vs $m_{D,3}$, with size determined by $\Delta m^2_{21} \approx 4\,\varepsilon \times m_{atm}^2$, giving $\varepsilon \approx 0.0074$. This is well within the bounded-spurion principle of Paper 23 Appendix B ($\|\varepsilon\| \lesssim A \approx 0.080$), at $\varepsilon \approx A / 11$. The two breakings (Majorana exact, Yukawa ε) have independent physical origins. Restated gate: if a first-principles derivation of $\varepsilon \approx 0.0074$ from the (N_2, N_3) Yukawa structure (ZS-M11 lepton-sector D_5-ρ_4 channel under Z_2-breaking, cf. §9.5 of ZS-M11 v1.0 April 2026 update) cannot be obtained within ten years, the IO interpretation of ZS-S2 must be reconsidered as a phenomenological success without microphysical justification for the solar splitting. Status: OPEN. Anti-numerology note: the observed $\varepsilon \approx 0.0074$ is numerically close to $A / Q = (35/437) / 11 \approx 0.00728$ (1.6% mismatch); this is an OBSERVATION only and is NOT a claim, pending a derivation chain. A / Q is the natural Schur-complement coupling appearing in The Book §G.2 T1-2 fine-structure constant conjecture ($1/\alpha_{EM} \approx Q / A = 137.343$), so a future derivation linking the two channels through the same A / Q coupling is conceivable but currently unproven. Note: F-S2-IO3 falsifies the unification conjecture only, NOT the IO assignment itself; the IO assignment of §3.1 survives via the corrected Yukawa-side picture above.

§9. Conclusion

This paper establishes the complete neutrino sector prediction of Z-Spin Cosmology from the single geometric impedance A = 35/437, with zero new free parameters. The Type-I seesaw with m_D = m_e × A = 40.93 keV yields an HNL at M_R = 33.50 GeV with mixing angle $|\theta|^2 = 1.49 \times 10^{-12}$ — a "33 GeV Ghost" that is 10^6–10^7 times below all current and projected direct search bounds. The HNL lifetime τ = 38 ns ensures complete BBN safety. The Z_2 flavor symmetry from $\hat{W}^2=I$ forces tree-level quasi-degeneracy $M_2=M_3$, motivating μ–τ reflection symmetry ($\theta_{23}=45°$, δ_CP=±π/2) and establishing the structural prerequisite for resonant leptogenesis (ZS-S5 v1.0). Davidson–Ibarra exclusion rules out hierarchical leptogenesis, leaving only resonant/ARS mechanisms compatible with the Z-Spin mass spectrum. Five falsification gates (§8) are pre-registered.

§10. Verification Suite (25/25 PASS)

Category	Tests	Pass/Fail	Key Result
Seesaw Mass Hierarchy	5	5/0	m_D, M_R, Y_0^2, m_ν closure, hierarchy
HNL Invisibility	5	5/0	$|\theta|^2 = 1.49\times10^{-12}$, below all bounds, LEP safe
BBN Safety	5	5/0	τ=38 ns, Boltzmann suppression, M_R < T_sph
Z_2 Flavor & μ-τ	5	5/0	Z_2 involution, ΔM hierarchy, |sinδ|=1 chain
Resonance Gap & Falsification	5	5/0	DI bound, ΔM/Γ, ARS compatible, 5 gates
TOTAL	**25**	**25/0**	**100% pass rate**

§10.1 IO Consistency Test Extension (April 2026)

The 25/25 PASS verification of ZS-S2 v1.0 (above) is preserved unchanged. The April 2026 update addenda (§3.1, §6 N_1 identification and character orthogonality, §8.1) introduce no new numerical claims beyond what is already verified — they are interpretive and structural extensions. However, the following 6 new test items are registered for the next verification script update (ZS_S2_verify_v1_0_apr2026.py, status: PENDING). Each new test is a numerical or structural cross-check that does not affect the existing 25 tests:

T26 (PENDING): Verify U = diag(+1, −1, −1) is a valid representation of $\hat{W}^2=I$ on the 3-generation HNL space with eigenvalues (+1, −1, −1) — group-theoretic check, expected PROVEN.

T27 (PENDING): Verify NO ↔ IO assignment consistency: under both interpretations, $m_D^2/M_R = \sqrt{|\Delta m^2_{31}|} \approx 0.050$ eV is satisfied identically; only the assignment of N_i to m_i changes.

T28 (PENDING): Compute Σm_ν^IO = $m_1 + m_2 + m_3$ with $m_1 = \sqrt{|\Delta m^2_{31}|}$, $m_2 = \sqrt{(|\Delta m^2_{31}| + \Delta m^2_{21})}$, $m_3 = 0$. Expected: 0.0992 eV. Compare against Planck18+BAO bound 0.12 eV (PASS expected) and DESI strict 0.071 eV (FAIL expected; tension acknowledged).

T29 (PENDING): F-Q5.8 (ZS-Q5 v1.0 §9) cross-link consistency: verify that the IO statement in §3.1 of this paper is byte-equivalent to the IO statement in ZS-Q5 v1.0 §4.5 within stated tolerances.

T30 (PENDING): Compute the IO m_ββ envelope for all Majorana-phase combinations, with Z-Spin PMNS parameters (θ_{12}, θ_{13}, θ_{23}) and $m_3 = 0$. Expected: m_ββ ∈ [0.0152, 0.0496] eV. Compare against KamLAND-Zen bound 0.036 eV; verify partial PASS region exists.

T31 (PENDING, April 2026 update): Verify dim Hom_I(1, 3 ⊗ 5 ⊗ 1) = ⟨χ_3, χ_5⟩ = 0 by direct character orthogonality computation using the explicit $I \cong A_5$ character table (60-element class sum). Expected: exact zero by construction. Cross-check: compute dim Hom_I(1, 3 ⊗ 5 ⊗ X) for X ∈ {1, 3, 3′, 4, 5} and verify the result vector is (0, 1, 1, 1, 1), confirming that the trivial irrep 1 is uniquely the irrep that forbids the Yukawa coupling. This test is the ZS-S2-side companion of ZS-M11 v1.0 §9.5.4 T25 (April 2026 update). Expected PASS by exact arithmetic.

April 2026 update verification status: 25/25 PASS (v1.0 inherited, unchanged) + 6 PENDING (T26–T31, awaiting ZS_S2_verify_v1_0_apr2026.py update; T31 is the April 2026 update addition for character orthogonality cross-check with ZS-M11 v1.0 §9.5.4 T25). Target after script update: 31/31 PASS.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

Verification script: ZS_S2_verify_v1_0.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_S2_verify_v1_0.py. Expected output: 25/25 PASS, exit code 0. The verification suite is publicly available.

Appendix A: Cross-Reference Table

Paper	Used In	Direction	Relation
ZS-F2 v1.0	A = 35/437	Input → ZS-S2	LOCKED
ZS-F5 v1.0	$\hat{W}^2$=I, κ=4 Z_2	Input → ZS-S2 §6	PROVEN
ZS-M2 v1.0	m_D = m_e×A	Input → ZS-S2 §3	DERIVED
ZS-S1 v1.0	U(1)_Y = Z-sector	Input → ZS-S2 §3	DERIVED
ZS-S4 v1.0	Texture Zero Lemma	→ ZS-S2 §7.2 (ref)	PROVEN
ZS-S5 v1.0	Leptogenesis framework	ZS-S2 → ZS-S5 (exports)	DOWNSTREAM
ZS-U7 v1.0	QKE kernel closure	ZS-S2 → ZS-U7 App.A	DOWNSTREAM
ZS-U4 v1.0	Global fit, BBN	ZS-S2 → ZS-U4 §7	DOWNSTREAM

cross-reference additions (April 2026): Two additional cross-paper relations are registered for the IO assignment introduced in §3.1 and §6:

• ZS-Q5 v1.0 §4.1a (contragredient branch) → ZS-S2 §3.1: provides the contragredient branch derivation that uniquely selects $\delta_CP = -\pi/2 - \arctan(A) = 265.42°$ and Inverted Ordering as the Z-Spin canonical mass ordering. Relation: INPUT (DERIVED-CONDITIONAL on NC-Q5.2). Cross-link: gate F-Q5.8 (ZS-Q5 v1.0 §9) is registered as F-S2-IO0 in §8.1.

• The Book §7.3, §27.4 → ZS-S2 §6: provides the explicit U = diag(+1, −1, −1) seam involution and the "N_1 Z_2-singlet" identification. Relation: INPUT (PROVEN). The Book §7.3 wording ("leaves N_1 invariant and exchanges the signs of N_2 and N_3") is the canonical statement; ZS-S2 §6 imports it explicitly to make the tree-level group structure unambiguous in the canonical neutrino sector source paper.

Note: Prior to , ZS-S2 v1.0 was internally consistent with itself but was not synchronised with ZS-Q5 v1.0 §4.1a or The Book §27.4 on the IO selection. The update closes this cross-paper consistency gap without altering any existing numerical claim.

Appendix B: Derivation Chain Summary

#	Statement	Source	Status
1	A = 35/437	ZS-F2 v1.0	LOCKED
2	$m_D = m_e \times A = 40.93$ keV	ZS-M2 v1.0	DERIVED
3	$m_atm = 0.050$ eV	NuFIT 5.2	STANDARD
4	$M_R = m_D^2/m_atm = 33.50$ GeV	Type-I seesaw	DERIVED
5	$\|\theta\|^2 = (m_D/M_R)^2 = 1.49\times10^{-12}$	Steps 2+4	DERIVED
6	$\tau_N = 38$ ns	Eq.(7–8)	DERIVED
7	$\hat{W}^2=I \rightarrow M_2=M_3$	ZS-F5 v1.0	PROVEN
8	DI bound excludes hierarchical	Davidson–Ibarra	DERIVED

derivation chain extension (April 2026):

9. U = diag(+1, −1, −1) on (N_1, N_2, N_3) — N_1 Z_2-singlet, (N_2, N_3) Z_2-doublet. Source: $\hat{W}^2=I$ (ZS-F5 v1.0) → κ=4 witness → 3-generation projection. Status: PROVEN. Cross-link: The Book §7.3.

10. Inverted Ordering canonical assignment: (N_2, N_3) → (m_1, m_2) at $\sqrt{|\Delta m^2_{31}|} \approx 0.05$ eV; $N_1 \rightarrow m_3 \approx 0$. Source: Step 9 (PROVEN) + ZS-Q5 v1.0 §4.1a contragredient branch (DERIVED-CONDITIONAL on NC-Q5.2). Status: DERIVED-CONDITIONAL. Cross-link: ZS-S2 §3.1, ZS-Q5 v1.0 §4.5, The Book §27.4.

11. $\Sigma m_\nu^{IO} = m_1 + m_2 + m_3 \approx 0.0992$ eV ($m_3 \approx 0$). Source: Step 10 + NuFIT 6.0 mass-squared splittings. Status: TESTABLE (gate F-S2-IO1, §8.1). Cross-link: ZS-U4 v1.0 (global cosmological fit, IO sync TODO).

12. $m_\{D,1\} = 0$ (N_1 decoupled from SM lepton doublets, "minimal seesaw"). Source: Step 9 (PROVEN) + ZS-M9 v1.0 §3 Table 2 assignment ν_R ↔ I-irrep 1 (HYPOTHESIS strong) + ZS-M11 v1.0 §9.5.1 Theorem (April 2026 update): $\dim \mathrm{Hom}_I(1, 3 \otimes 5 \otimes 1) = \langle \chi_3, \chi_5 \rangle = 0$ by character orthogonality (PROVEN). Status: DERIVED-CONDITIONAL (April 2026 update; upgraded from HYPOTHESIS).

Cross-link: ZS-S2 §6 (April 2026 update), ZS-M11 v1.0 §9.5 (April 2026 update), ZS-M9 v1.0 §3 (April 2026 update), Frampton–Glashow–Yanagida 2002.

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[25] K. Kang, "McKay Correspondence and SM Field Classification," ZS-M9 v1.0 (2026). [§3 Table 2 SM field assignment, including ν_R ↔ I-irrep 1; April 2026 update adds the downstream consequence note]. (April 2026 addition)

[26] K. Kang, "Icosahedral Yukawa Completion," ZS-M11 v1.0 (2026). [§9.5 Lepton Sector: Singlet ν_R Yukawa Vanishing, Theorem 9.5.1 dim Hom_I(1, 3 ⊗ 5 ⊗ 1) = 0 by character orthogonality; April 2026 update]. (April 2026 addition)

Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.1.0.)

Type-I seesaw with m_D = m_e × A = 40.93 keV predicts HNL at M_R = 33.50 GeV. Seesaw mixing $|\theta|^2 = 1.49 \times 10^{-12}$ is 10^6–10^7 below all bounds. HNL lifetime τ = 38 ns ≪ t_BBN (BBN safe). Z_2 flavor symmetry from $\hat{W}^2=I$ forces $M_2=M_3$ tree-level degeneracy. Davidson–Ibarra exclusion confirms hierarchical leptogenesis impossible; only resonant/ARS compatible. CANONICAL source for all neutrino sector parameters. Verification: 25/25 PASS. Zero free parameters.

Z-Sim cross-reference (March 2026): All 8 closure parameters of the Z-Spin forward simulator are now DERIVED from A = 35/437 and (Z,X,Y) = (2,3,6). See ZS-Q7 v1.0 §5.8, ZS-M3 v1.0 §12, ZS-T3 v1.0. Zero free parameters.

(April 2026):v1.0 — April 2026 update: Cross-paper IO consistency synchronisation with ZS-Q5 v1.0 §4.1a, The Book §27.4, ZS-M9 v1.0 §3 Table 2, and ZS-M11 v1.0 §9.5. No prior content removed; all v1.0 numerical claims preserved unchanged. External label remains v1.0 (no version bump, no citation cascade across the corpus). Additions: (a) §2 Locked Inputs Note on m_atm extended to clarify that $|\Delta m^2_{31}|$ is NO/IO-independent. (b) NEW §3.1 Mass Ordering Assignment establishes the canonical IO assignment (N_2, N_3) → (m_1, m_2) at the atmospheric scale and $N_1 \to m_3 \approx 0$, with cosmological prediction $\Sigma m_\nu \approx 0.0992$ eV. STATUS: DERIVED-CONDITIONAL on ZS-Q5 v1.0 §4.1a contragredient branch. (c) §6 extended with explicit U = diag(+1, −1, −1) seam involution and N_1 Z_2-singlet identification (PROVEN, importing The Book §7.3). N_1 Yukawa structure m_{D,1} = 0 now registered as DERIVED-CONDITIONAL (upgraded from HYPOTHESIS) via the character orthogonality chain dim Hom_I(1, 3 ⊗ 5 ⊗ 1) = ⟨χ_3, χ_5⟩ = 0 (PROVEN at character level by ZS-M11 v1.0 §9.5.1, conditional only on the ZS-M9 v1.0 §3 Table 2 assignment ν_R ↔ I-irrep 1). (d) NEW §8.1 registers four IO consistency gates: F-S2-IO0 (cross-link to ZS-Q5 F-Q5.8, JUNO/DUNE mass ordering), F-S2-IO1 (Σm_ν cosmological), F-S2-IO2 (m_ββ from 0νββ), F-S2-IO3 (theoretical: Yukawa-side spurion derivation, OPEN). (e) NEW §10.1 registers six new verification tests T26–T31 as PENDING (awaiting ZS_S2_verify_v1_0_apr2026.py update; target 31/31 PASS). (f) Appendix A cross-reference table extended with ZS-Q5 v1.0 and The Book entries. (g) Appendix B derivation chain extended with steps 9–12; step 12 (m_{D,1} = 0) upgraded from HYPOTHESIS to DERIVED-CONDITIONAL. (h) References [21] NuFIT 6.0 (Esteban et al. 2024), [22] ZS-Q5 v1.0, [23] KamLAND-Zen 2023, [24] Frampton-Glashow-Yanagida 2002 added. (i) F-S2-IO3 rewritten: the original conjecture that the leptogenesis resonance gap and the IO solar splitting share a common Majorana-side spurion is FALSIFIED at order-of-magnitude (the ratio ΔM_pair / ΔM_resonance ≈ 10^{15} cannot be reconciled). The corrected picture uses an exact Z_2-degenerate Majorana side (preserving leptogenesis resonance) plus a small Yukawa-side spurion ε ≈ 0.0074 ≈ A/11 (well within bounded-spurion ‖ε‖ ≲ A ≈ 0.080); the two breakings have independent physical origins. The numerical proximity ε ≈ A/Q is registered as an OBSERVATION only (not a claim), pending derivation. (j) T31 character orthogonality verification test added in §10.1, paired with ZS-M11 v1.0 §9.5.4 T25. (k) References [25] ZS-M9 v1.0 and [26] ZS-M11 v1.0 added, formalising the cross-paper derivation chain for m_{D,1} = 0. The 25/25 v1.0 verification suite is preserved unchanged; April 2026 update verification status is 25/25 PASS + 6 PENDING (T26–T31). Zero new free parameters; A = 35/437 remains the sole geometric input. The IO assignment is DERIVED-CONDITIONAL and is in mild tension (~2.7σ) with the current NuFIT 6.0 NO preference; resolution expected from JUNO (~2027), DUNE (~2030), DESI DR3, and LEGEND-1000 (~2030). This update is part of a coordinated four-document batch synchronising ZS-S2, ZS-M9, ZS-M11, and The Book §7.3 in a single April 2026 release.